

MIKE E. UNDERWOOD

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Summary

Principal-level software engineer with a strong foundation in full-stack development, distributed systems, and cloud-native engineering. Expert in .NET, C#, Java, SQL, React/TypeScript, and event-driven architectures using Azure and AWS Lambda. Experienced in building scalable APIs, payment workflows, and high-traffic digital experiences while applying deep debugging, performance tuning, and observability practices. Known for pairing hands-on implementation with strong system-design skills to modernize platforms, reduce operational complexity, and deliver reliable, production-ready solutions. Additional writing, system-design thinking, and engineering projects are available at mikeeunderwood.dev.

Skills & Technical Expertise

Core Languages

- C#, Java, C++, JavaScript/TypeScript, HTML5
- Python (automation, scripting, test harnesses)
- SQL (T-SQL, MySQL/MariaDB SQL, stored procedures, query optimization)

Frameworks, Tools & Platforms

- .NET, ASP.NET Core (REST APIs, MVC, middleware), ASP.NET MVC (CMS UI components, Razor views)
- React, Vue.js, Node.js
- Sitecore, Optimizely (enterprise CMS)
- PowerShell, Postman, NUnit, xUnit

Cloud & Infrastructure

- Azure, AWS Lambda (serverless execution, event-driven workflows), Azure DevOps
- CI/CD automation (Azure Pipelines, GitHub Actions)
- Docker, Kubernetes, IIS, Windows, Red Hat Linux
- Observability & distributed tracing (Azure Application Insights, logging, telemetry)

Data Platforms

- SQL Server, Azure SQL Database, MySQL/MariaDB
- Database modernization & migration (on-prem → cloud native)

Architecture & Engineering Practices

- REST/JSON API design, distributed systems, microservices, event-driven workflows (Azure Service Bus)
- Compliance-driven engineering (PCI, HIPAA), secure integrations, transaction-workflow automation
- Legacy modernization → cloud-native architectures
- Agile/SDLC (Azure DevOps, JIRA), DevOps practices

Reliability & Tooling

- Git, automated testing (unit, integration, API validation)
- Debugging, performance tuning, scalability engineering
- SRE principles: release reliability, monitoring, incident response

Experience

Rightpoint — Remote, United States

Solutions Architect (Jan 2020 – Nov 2025)

- Engineered CMS presentation-layer components using ASP.NET MVC, Razor views, controllers, and view models; built headless front-end experiences in React/TypeScript consuming C#/.NET APIs.
- Delivered full-stack features across React, Vue.js, Node.js, SQL Server, and Azure SQL, supporting high-traffic retail/e-commerce workloads and personalized digital experiences.
- Designed and implemented payment-processing workflows using C#/.NET Web APIs and AWS Lambda for serverless execution, publishing lifecycle events into Azure Service Bus to support asynchronous, event-driven processing.
- Built and deployed cloud-native applications using Azure Pipelines targeting AKS, Azure VMs, and AWS Lambda-based serverless components; integrated Application Insights for distributed tracing and real-time diagnostics.
- Analyzed logs, metrics, and distributed traces to identify performance bottlenecks and reliability issues; implemented fixes across API layers, data-access patterns, and cloud infrastructure.
- Authored REST API specifications (JSON/YAML) with sample code and integration patterns; implemented secure integrations with HIPAA-compliant third-party systems for healthcare and retail clients.

- Troubleshoot production issues across front-end, API, database, and cloud layers, delivering iterative improvements aligned with SRE principles for reliability, scalability, and maintainability.

AXIS Capital — Alpharetta, GA

Application Delivery Lead, Corporate Systems (Mar 2018 – Dec 2019)

- Designed and implemented backend components and integration workflows for corporate systems using C#/.NET, SQL, and automation scripts, translating business requirements into reliable, production-ready services.
- Built and enhanced internal REST APIs and service interfaces aligned with enterprise architecture standards, improving interoperability across financial, HR, and compliance platforms.
- Developed technical designs, sequence flows, and implementation patterns for new features; contributed hands-on to coding, debugging, and integration testing across multiple applications.
- Improved system performance and reliability by profiling services, optimizing data-access logic, and resolving defects across API, database, and infrastructure layers.
- Collaborated with offshore DEV/QA teams through code reviews, implementation patterns, and technical guidance to ensure consistent engineering quality.
- Supported production systems by analyzing logs and telemetry, troubleshooting issues across distributed components, and implementing fixes to stabilize critical business workflows.
- Worked with infrastructure teams to deploy application updates, validate environment configurations, and ensure smooth release cycles across development, QA, and production environments.

Fiserv — Alpharetta, GA

Dec 1999 – Feb 2018

Role progression: Senior Software Engineer → Software Architect → Senior Software Architect → Manager & Lead Senior Software Architect → Principal Solutions Architect

Principal Solutions Architect (Aug 2014 – Feb 2018)

- Engineered core components for a multi-year migration of walk-in payment applications from legacy Informix/Unix to C#, PowerShell, and SQL Server, delivering modernized services, infrastructure, and deployment automation.
- Built .NET Web APIs enabling barcode-based walk-in payments at retail POS, eliminating manual entry errors and improving transaction accuracy across high-volume payment networks.
- Refactored and optimized compliance-reporting pipelines, achieving 100% reporting accuracy across all walk-in payment transactions through improved data validation, ETL logic, and error-handling patterns.
- Led hands-on development and troubleshooting during production deployments, resolving high-severity issues across application, database, and network layers to stabilize mission-critical payment workflows.
- Designed and implemented reusable technical patterns and component architectures supporting long-term roadmap initiatives for the CheckFreePay division.
- Collaborated with product, engineering, and support teams to translate business requirements into implementable technical designs, contributing directly to coding, debugging, and integration testing.

Manager & Lead Senior Software Architect (Mar 2012 – Jul 2014)

- Designed and implemented a unified .NET Web API layer across Biller Solutions products (spanning .NET and Java backends), enabling consistent integration patterns and reducing onboarding friction for new features.
- Engineered platform enhancements aligned to multi-year roadmap initiatives, contributing directly to service design, API contracts, data-access logic, and integration workflows.
- Delivered hands-on development across multiple services, implementing new features, debugging production issues, and optimizing performance across API and database layers.
- Built reusable implementation patterns and code frameworks that improved engineering consistency and accelerated delivery across teams.
- Performed deep-dive troubleshooting across distributed systems, resolving defects in authentication flows, service orchestration, and data-exchange logic.
- Participated in architecture reviews while remaining directly involved in coding, refactoring, and technical decision-making for high-impact features.

Senior Software Architect (Sep 2008 – Feb 2012)

- Designed and implemented the eBill Easy Activation platform (patent-pending), increasing e-bill activations by 230% through automated enrollment driven by data-warehouse logic and optimized decision workflows.
- Built a unified authentication mechanism using encryption + hashing, reducing digital certificates from hundreds to a single managed certificate and significantly lowering operational overhead for Fiserv and biller clients.
- Engineered a SaaS eBill presentment platform using .NET APIs adopted by major banks; this system became the core technology for Fiserv's eBill presentment solution used by 4,000+ financial institutions.
- Designed and implemented the responsive iOS/Android mobile version of MyCheckFree.com, including Java REST APIs, JSON payloads, and a scalable API-versioning strategy.

- Built the .NET e-bill platform that reduced implementation timelines from nine months to two, establishing the foundation for Fiserv's eBill Distribution™ product still in active commercial use.
- Delivered hands-on engineering across .NET, Java, and SQL layers, implementing new features, debugging production issues, and optimizing performance across distributed payment and presentment systems.

Software Architect (Jul 2001 – Aug 2008)

- Designed and implemented a Java-based biller-direct payment and presentment platform integrating iSeries toolkits, mainframe payment systems, and HTML widgets; built a polymorphic API layer to support biller-specific logic and reduce onboarding complexity.
- Engineered multiple ASP.NET payment platforms (retail, apartment communities, charities), serving as team lead for ACH engines, email systems, and integration workflows; delivered production-ready code across C#, .NET, SQL, and C++.
- Developed a Tandem OS point-of-sale solution in C and built a C++-based Tandem emulator to accelerate early-stage development and reduce dependency on limited hardware environments.
- Migrated BlueGill Technologies' iSeries eBill/payment toolkit from Toronto to Norcross as part of the acquisition integration effort, modernizing components and ensuring compatibility with Fiserv's broader payment ecosystem.
- Contributed to the design and implementation of a .NET-based SOA layer exposing mainframe capabilities via utility, domain, channel, and orchestration services; served as lead engineer for eBill-related service components.
- Delivered hands-on engineering across distributed systems, implementing new features, debugging complex integration issues, and optimizing performance across API, data, and legacy system boundaries.

Senior Software Engineer (Dec 1999 – Jun 2001)

- Designed and implemented client-integration solutions ranging from SSO flows to full hosted payment experiences using CheckFree's API interfaces, delivering reliable, production-ready code across C#, .NET, and SQL.
- Built custom authentication, data-exchange, and payment-processing components for enterprise clients, ensuring secure interoperability with existing financial systems.
- Developed technical estimates, implementation plans, and integration deliverables while contributing directly to coding, debugging, and end-to-end testing.
- Engineered reusable integration patterns that reduced onboarding time for new clients and improved consistency across multiple payment-processing implementations.
- Troubleshot production issues across API, data, and network layers, implementing fixes that stabilized high-volume payment workflows.

RIB Software (formerly Building Systems Design) — Atlanta, GA

Senior Software Engineer (Jul 1991 – Nov 1999)

- Designed and implemented cost-estimation and lifecycle-management capabilities used by the U.S. Army Corps of Engineers, improving project accuracy and analytical decision workflows through optimized algorithms and data-modeling techniques.
- Co-architected and developed BSD SpecLink, an automated building-specification platform built in C++, SQL, and systems-integration layers; delivered core features that enabled adoption across 1,000+ construction projects and remain in active commercial use.
- Refactored and modernized core product components to improve scalability, reduce long-standing technical debt, and streamline release cycles across multiple product generations.
- Engineered enterprise-grade features across C++, SQL, and integration layers, improving system reliability, performance, and maintainability for large-scale construction-specification workflows.
- Delivered full-lifecycle development — requirements, data modeling, coding, debugging, and deployment — ensuring stable, production-ready releases across diverse customer environments.

Sales Technologies — Buckhead, GA

Software Engineer (Jul 1990 – Jun 1991)

- Engineered modular C++ components for a sales-force automation platform, improving scalability, strengthening database interoperability, and delivering reliable, production-ready features that supported increased concurrent-user demand.

Universal Microsystems — Peachtree Corners, GA

Software Engineer (Jul 1987 – Jun 1990)

- Designed and delivered custom enterprise software solutions for clients including Coca-Cola, Siemens, and Digital Communications Associates, engineering full-lifecycle systems across Clipper, early Windows platforms, and multiple databases to automate high-volume business processes and improve operational reliability.

Education

Master of Science: Computer Information Systems

Georgia State University: J. Mack Robinson College of Business

1989

Atlanta, Georgia, United States

BBA: Computer Information Systems

Georgia State University: J. Mack Robinson College of Business

1986

Atlanta, Georgia, United States.

Certifications

Sitecore Experience Solution 9 Developer (Aug 2020)

LinkedIn Learning: Cloud Architect Core Concepts, Creating Your IT Strategy (2018)